Near Final Project Review

NAME:- SREEJA SANYAL

WEEK :-4

DAY:-27

DATE:- 30.10.2026

Final Methodology

- *Collected and processed lithium-ion battery data*
- *Selected relevant features such as Voltage, Current, Temperature, Elapsed Time, Cycle and Power*
- *Prepared data for model training and evaluation*
- *Selected XGBoost Regressor for SOC estimation*
- *Selected Random Forest Regressor for Remaining Charging Time prediction*
- *Evaluated model performance using MAE, RMSE and R^2*

Results Summary

- *Final SOC model: XGBoost Regressor*
- *Final Remaining Charging Time model: Random Forest Regressor*
- *Evaluated using validation and test metrics*
- *Reviewed prediction performance and model errors*
- *Results checked against available evaluation outputs*
- *Final reported metrics retained consistently*

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FINAL MODEL RESULT SUMMARY
SOC Prediction
Model: XGBoost Regressor
Test R²: 0.990521
Remaining Charging Time Prediction
Model: Random Forest Regressor
Test R²: 0.998756
Remaining Charging Time Validation
Model: Random Forest Regressor
Validation R²: 0.996502
Final Feature Set
1. Voltage
2. Current
3. Temperature
4. Elapsed Time
5. Cycle
6. WhAccu
7. Power
Important:
Capacity is not an input feature of the frozen final models.
Model Selection
SOC:
XGBoost Regressor
Remaining Charging Time:
Random Forest Regressor
The reported values are based on the existing frozen
validation/test evidence. No retraining was performed during
Day 26.
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Software Artefacts

- Google Colab notebooks
- Processed battery dataset
- Saved SOC estimation model
- Saved Remaining Charging Time model
- Model evaluation outputs
- Supporting plots and result tables

Reproducibility

- Reviewed the project workflow and notebook sequence
- Checked data-processing and feature-selection steps
- Reviewed model training and evaluation procedures
- Final result table created
- Feature list verified
- Checked saved model files and output locations

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DAY 26 SUBMISSION NOTE
Day 26 software packaging and documentation completed.
Completed:
[✓] Frozen SOC model packaged
[✓] Frozen Remaining Charging Time model packaged
[✓] Actual model feature list verified
[✓] Inference workflow created
[✓] Example input created
[✓] Results table created
[✓] README created
[✓] Final report draft created
[✓] Validation records preserved
[✓] Day 25 analysis preserved
[✓] Software package ZIP created
Final feature list:
Voltage
Current
Temperature
Elapsed Time
Cycle
WhAccu
Power
No retraining was performed during Day 26.
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Limitations

- Prediction errors may occur under certain charging conditions
- Model performance may vary across temperatures and operating conditions
- Unusual battery behaviour may affect prediction accuracy
- Performance depends on the quality and coverage of the available dataset
- Further improvements may be possible through additional testing and analysis

Conclusion

The project successfully implemented XGBoost and Random Forest models for battery SOC estimation and Remaining Charging Time prediction. The models demonstrated predictive performance on the available dataset.

Thanks!